Selective Underwater Image Enhancement for Object Detection with YOLO11

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Abstract*—Underwater image enhancement (UIE) is widely assumed to benefit object detection, yet applying SCNet or Semi-UIR uniformly lowers YOLO11n mAP@0.5:0.95 on the 4,200-image RUOD benchmark (from 0.6155 for raw images to 0.6129 and 0.6084, respectively). A ground-truth-guided oracle that picks the best version per image reaches 0.6296 (+0.0142 dataset mAP, +0.0231 per-image mAP), but whether this gap can be captured without labels has remained an open question. We design a leakage-free selective enhancement pipeline: out-of-fold labels are generated using a three-fold detector evaluation protocol with an improvement margin ($\delta = 0.01$), and an ultra-lightweight ResNet18 selector (7.7 ms latency) dynamically chooses among raw, SCNet, and Semi-UIR inputs based solely on the raw image. With decision thresholding ($\tau = 0.625$), the selector achieves 0.6159 mAP@0.5:0.95, matching the raw baseline (+3.2% dataset oracle gap closed) while safely keeping the raw image for 85.5% of queries. Crucially, a diagnostic noise floor analysis reveals that an oracle over the raw image and its horizontal flip—involving zero enhancement—achieves +0.0207 per-image mAP gain (144% of SCNet's oracle gain of +0.0144), proving that the apparent oracle headroom is predominantly detector measurement stochasticity rather than learnable optical features. Finally, we evaluate actionable alternatives: multi-view Weighted Boxes Fusion (WBF) across spatial and photometric domains reliably improves detection (+0.0023 mAP@0.5:0.95, 95% CI $[+0.0014, +0.0051]$), and multi-version detector training boosts raw-image mAP@0.5:0.95 from 0.6037 to 0.6210 (+0.018), confirming that UIE provides genuine value as training-time data augmentation rather than inference-time image selection.

***Index Terms*—underwater image enhancement, underwater object detection, selective enhancement, YOLO11, RUOD, UIEB**

I. Introduction

Underwater optical vision systems deployed on autonomous underwater vehicles (AUVs) and remotely operated vehicles (ROVs) encounter severe physical degradations, including wavelength-dependent light absorption and particulate scattering [1, 2]. Red light attenuates rapidly within shallow depths, producing pervasive green-blue color casts, low contrast, boundary blurring, and turbidity haze.

To restore visual clarity, deep learning underwater image enhancement (UIE) models such as SCNet [3] and Semi-UIR [4] have demonstrated remarkable success according to full-reference perceptual metrics such as PSNR and SSIM. Consequently, a prevailing engineering assumption in marine robotics is that perceptual restoration should serve as an automatic, modular pre-processing stage preceding downstream object detectors.



Fig. 1. Sample RUOD image before and after enhancement. From left to right: raw input (orig), Semi-UIR [4], and SCNet [3].

However, recent studies reveal a fundamental paradox [5]: visual quality does not consistently translate to downstream detection accuracy. Uniform enhancement across diverse benchmarks frequently degrades detection performance relative to processing raw images directly. Enhancers can introduce artificial high-frequency textures, amplify sensor noise, oversaturate color channels, or alter fine object contours. Nonetheless, enhancement remains beneficial on select severely degraded, turbid, or under-illuminated inputs.

Awad et al. [5] highlighted this heterogeneity on the RUOD benchmark [1] via a theoretical "mixed-set upper bound": an idealized oracle selector picking the highest-mAP version among nine candidate enhancers increased mean per-image mAP from 0.68 to 0.77 (+0.09). This observation motivated two fundamental questions:

1) *Can selective enhancement be achieved autonomously?* An oracle requires ground-truth annotations at test time. Can a lightweight selector reliably choose the optimal image version using only the raw input?
2) *What is the true nature of the oracle headroom?* Awad et al. evaluated ten independent detectors across ten enhanced variants. Did the +0.09 headroom reflect recoverable visual signals, or was it largely an artifact of detector measurement noise and model diversity?

To address these questions, we implement and benchmark a selective enhancement pipeline for YOLO11n [6] on the 14,000-image RUOD benchmark. The pipeline selects among three versions: raw image, SCNet, and Semi-UIR. Our core contributions are:

- **Leakage-Free Selective Pipeline:** Out-of-fold selector targets are generated via a three-fold cross-validation

detector protocol with an improvement margin ($\delta = 0.01$), preventing in-sample overfitting. An ultra-lightweight ResNet18 selector [7] (7.7 ms latency) operates strictly on raw 224×224 inputs.

- **Safe Baseline Parity:** On 4,200 RUOD test images, the calibrated selector achieves 0.6159 mAP@0.5:0.95 (+3.2% dataset oracle gap closed), matching raw baseline accuracy (0.6155) while avoiding the degradation of uniform SCNet (0.6129), Semi-UIR (0.6084), or random selection (0.6120) by retaining the raw image for 85.5% of inputs.
- **Oracle Noise Floor Formulation:** Counter-experiments using horizontal reflection reveal that an oracle over raw and flipped images—involving zero enhancement—yields +0.0207 per-image mAP gain (144% of SCNet's gain of +0.0144). We prove that oracle headroom is predominantly detector measurement stochasticity rather than learnable optical features, explaining the selector's learning ceiling.
- **Actionable Alternatives:** Multi-view Weighted Boxes Fusion (WBF) [8] across geometric and photometric domains achieves statistically validated gains (+0.0023 mAP@0.5:0.95, 95% CI $[+0.0014, +0.0051]$). Furthermore, multi-version detector training boosts raw-image mAP@0.5:0.95 from 0.6037 to 0.6210 (+0.018), confirming that UIE provides genuine value as training-time data augmentation.

II. Related Work

A. Underwater Image Enhancement

Deep UIE models are typically categorized into supervised, zero-shot, and semi-supervised approaches. SCNet [3] targets water-type diversity by learning *water-type desensitized* representations: across U-Net feature scales, it decorrelates spatial activations via instance whitening and standardizes channel moments (mean and variance).

Semi-UIR [4] mitigates paired data scarcity using a mean-teacher framework with contrastive regularization. A *reliable bank* stores pseudo-labels ranked by no-reference image quality assessment (NR-IQA). At inference, an Attention-guided Illumination Modulation network (AIMnet) modulates features using an estimated multi-scale Gaussian luminance map.

Both enhancers were benchmarked on UIEB [2] using PSNR and SSIM, which evaluate full-reference pixel reconstruction rather than downstream detection utility.

B. Enhancement for Underwater Object Detection

The interaction between enhancement and detection is multifaceted. Fu et al. [1] introduced the large-scale RUOD dataset to benchmark underwater detection, noting that standard enhancements produced inconsistent effects across detector families. Luo et al. [9] developed MAS-YOLOv11 with multi-scale attention heads for cluttered aquatic environments, while Huang et al. [10] proposed AMIEOD to dynamically select low-illumination experts.

Awad et al. [5] systematically evaluated nine enhancers across modern detectors, finding that uniform enhancement failed to universally outperform raw imagery and proposing a per-image mixed-set upper bound. However, their upper bound required test-time ground truth and confounded model diversity by employing ten independently trained detector checkpoints.

C. Ensemble Box Fusion and Test-Time Augmentation

Test-Time Augmentation (TTA) aggregates predictions across transformed inputs. In contrast to Non-Maximum Suppression (NMS) which discards overlapping candidates, Weighted Boxes Fusion (WBF) [8] clusters overlapping bounding boxes across candidate views and computes confidence-weighted coordinate averages, enhancing consensus localization.

III. Methodology

A. System Overview

Let $x \in \mathbb{R}^{H \times W \times 3}$ denote an input underwater RGB image. We consider two frozen, author-pretrained enhancers: SCNet (E_{SC}) and Semi-UIR (E_{SU}). The candidate pool is:

$$\mathcal{V}(x) = \{\, v_0, v_1, v_2 \,\} = \{\, x,\ E_{\mathrm{SC}}(x),\ E_{\mathrm{SU}}(x) \,\}. \quad (1)$$

An adaptive selector f_θ inspects the raw image x to predict index $\hat{k} \in \{0, 1, 2\}$, routing $v_{\hat{k}}$ to a fine-tuned YOLO11n detector D.

B. Enhancers and Processing Pipeline

Both enhancers run at native resolution (640 px long edge) with author-released weights without task-specific fine-tuning:

- **SCNet:** We isolate the enhancement decoder branch, discarding the auxiliary VGG16 perceptual loss network. The input is reflect-padded to multiples of 8, and the output is cropped back and quantized to $[0, 255]$.
- **Semi-UIR:** The network computes an estimated luminance map $L(x)$ by convolving the luminance channel with Gaussian kernels at $\sigma \in \{15, 60, 90\}$, averaging logarithmic responses, and normalizing to $[0, 255]$. The input and luminance map are reflect-padded to multiples of 16. Semi-UIR requires approximately 2 s per image.

All enhanced images are saved as high-quality JPEGs (quality $= 95$) with identical bounding box labels.

C. Out-of-Fold (OOF) Selector Labeling

To prevent in-sample detector memorization from biasing selector labels, we employ a three-fold Out-of-Fold (OOF) protocol:

1) The 9,800 RUOD training images are partitioned into three folds $\mathcal{F}_0, \mathcal{F}_1, \mathcal{F}_2$ using random seed 0 (MD5 hash: `ea44b385539c`).
2) For each fold $k \in \{0, 1, 2\}$, a YOLO11n detector is trained for 30 epochs on all three versions of the remaining two folds ($\approx$ 19,600 images) using AdamW.
3) The final checkpoint (`last.pt`) evaluates all three versions of the held-out fold $\mathcal{F}_k$ (conf $= 0.001$, IoU $= 0.7$).
4) Using `pycocotools`, exact per-image mAP@0.5:0.95 values $(m_{i,0}, m_{i,1}, m_{i,2})$ are computed. Unannotated images receive 0.

With an improvement margin $\delta = 0.01$, target labels are assigned as:

$$y_i = \begin{cases} 0 & \text{if } \max_{k\in\{1,2\}} m_{i,k} - m_{i,0} < \delta, \\ \arg\max_k m_{i,k} & \text{otherwise.} \end{cases} \quad (2)$$

D. Adaptive Selector Architecture and Decision Rules

The selector is an ImageNet-pretrained ResNet18 [7] with a 3-way linear classification head. It inspects only the raw image resized to $224 \times 224 \times 3$, executing in 7.7 ms on an NVIDIA Tesla T4 GPU.

- **Class-Weighted Cross-Entropy (`ce`):**

$$\mathcal{L}_{\text{ce}} = -\sum_{c=0}^{2} w_c \cdot \mathbb{I}(y = c) \log p_c, \quad (3)$$

  where w_c represents inverse class-frequency weights.
- **mAP Regression (`reg`):** Directly predicts continuous mAP values $(\hat{m}_0, \hat{m}_1, \hat{m}_2)$ via Smooth L1 loss.

1) Decision Threshold (τ): To avoid false-positive enhancement triggers, candidate $k^* = \arg\max_{k\in\{1,2\}} p_k$ is selected only if $p_{k^*} - p_0 > \tau$; otherwise, the raw image is retained. For regression, the rule is $\hat{m}_{k^*} - \hat{m}_0 > \tau$.

2) Validation Selection Criterion: Because always-raw achieves 58.2% training accuracy and 62.4% test accuracy trivially, models and thresholds are selected via *chosen-mAP* on a held-out 20% validation split (1,960 images):

$$\text{chosen-mAP} = \frac{1}{N_{\text{val}}} \sum_{i=1}^{N_{\text{val}}} m_{i,\hat{y}_i}. \quad (4)$$

Optimal thresholds are $\tau = 0.625$ for `ce` and $\tau = 0.040$ for `reg`.

E. Zero-Leakage Composition Evaluation

On the 4,200 RUOD test images:

1) A unified detector (`full.pt`) is trained for 50 epochs on all three versions of the 9,800 training images (29,400 images total; training time: 10,732 s).
2) Test predictions across candidate versions are precomputed once. Any policy is evaluated by composing stored prediction lists per image and executing `pycocotools`.
3) Statistical significance is established via 1,000 paired bootstrap resamples of per-image mAP differences against raw baseline, yielding 95% confidence intervals and bootstrap p-values $p_{\text{boot}}(\Delta \le 0)$.
4) Headroom utilization is quantified via Gap Closed:

$$G = \frac{\text{mAP}_{\text{policy}} - \text{mAP}_{\text{orig}}}{\text{mAP}_{\text{oracle}} - \text{mAP}_{\text{orig}}} \times 100\%. \quad (5)$$

F. Oracle Noise Floor Formulation

To determine whether the oracle upper bound reflects optical signal recovery or detector measurement variance, we introduce a counter-experiment using horizontal reflection (T_{flip}).

For input x, predictions $D(T_{\text{flip}}(x))$ are inverted back to original image coordinates via $T_{\text{flip}}^{-1}(\hat{\mathcal{B}})$. Because $T_{\text{flip}}(x)$ is photometrically identical to x, any divergence between $m_{i,\text{orig}}$ and $m_{i,\text{flip}}$ arises entirely from detector spatial sensitivity and NMS stochasticity. The *Noise Oracle* is defined as:

$$\text{Oracle}_{\{\text{orig,flip}\}}(i) = \max\left(m_{i,\text{orig}},\ m_{i,\text{flip}}\right). \quad (6)$$

When $\text{Oracle}_{\{\text{orig,flip}\}}$ matches or exceeds the gain of $\text{Oracle}_{\{\text{orig},E(x)\}}$, the apparent headroom of selective enhancement is demonstrated to be an artifact of detector measurement noise.

G. Multi-View Weighted Boxes Fusion (WBF)

We ensemble candidate bounding boxes across raw, flipped, SCNet, and Semi-UIR views using Weighted Boxes Fusion [8] with fixed parameters: $\text{IoU}_{\text{thr}} = 0.6$, skip_box_thr $= 0.001$, and at most 150 boxes per candidate view.

IV. Experimental Setup

A. Datasets

UIEB [2]: Contains 890 paired real-world underwater images. We use the standard 90-image test set at native resolution to verify enhancer numerical fidelity.

RUOD [1]: Contains 14,000 underwater images (9,800 train / 4,200 test) across 10 marine categories: *holothurian, echinus, scallop, starfish, fish, corals, diver, cuttlefish, turtle, jellyfish*. Long edges are resized to 640 px.

B. Implementation Details

Experiments run on Kaggle using dual NVIDIA Tesla T4 GPUs with PyTorch 2.11 (CUDA 12.8) and Ultralytics 8.4.174. Detection uses YOLO11n (640×640 input, AdamW optimizer, batch size 32, fixed seed 0). ResNet18 is trained for 15 epochs with AdamW (learning rate 3×10^{-4}, weight decay 10^{-2}) on an 80/20 train/val split (7,840 train / 1,960 val).

C. Metrics

Perceptual fidelity is measured via PSNR and SSIM using `scikit-image`. Detection accuracy is assessed using COCO-standard mAP@0.5 and mAP@0.5:0.95 via `pycocotools`. We report paired bootstrap 95% confidence intervals (CI) over 1,000 resamples.

V. Results and Discussion

A. Enhancer Verification on UIEB

Table I compares our measurements on the 90-image UIEB test split against published numbers. SCNet is within 0.35 dB PSNR and Semi-UIR is within 0.38 dB PSNR of published scores, with SSIM within 0.02. Both enhancers raise PSNR by over 4.5 dB over raw inputs.

Table II shows winning counts per image on UIEB. Even under full-reference fidelity, no single enhancer dominates across all samples: SCNet wins on 24 images and raw images win on 6 images. A per-image oracle reaches 24.95 dB (+0.74 dB over Semi-UIR). On image 703, raw PSNR is 40.63 dB, whereas SCNet and Semi-UIR drop to 27.54 dB and 26.38 dB. Conversely, on image 713, raw PSNR is 9.75 dB while SCNet and Semi-UIR reach 19.64 dB and 28.54 dB.

TABLE I
PSNR AND SSIM ON UIEB TEST SET (90 IMAGES, MEAN ± STD).

Method	Ours		Published	
	PSNR (dB)	SSIM	PSNR	SSIM
Raw input	17.90 ± 5.35	0.776 ± 0.137	–	–
SCNet [3]	22.43 ± 4.43	0.879 ± 0.082	22.08	0.8625
Semi-UIR [4]	$\mathbf{24.21} \pm 4.03$	$\mathbf{0.886} \pm 0.069$	24.59	0.9010
Per-image oracle†	24.95	0.9010	–	–

†Best of the three versions per image, chosen per metric.

TABLE II
NUMBER OF UIEB TEST IMAGES (OUT OF 90) WINNING PER METRIC.

Metric	Raw input	SCNet	Semi-UIR
PSNR	6	24	60
SSIM	10	29	51

B. Out-of-Fold Labeling on RUOD Training Set

On the 9,800 RUOD training images, the 3-fold OOF evaluation yielded:

- **Raw images (`orig`):** 5,704 images (**58.2%**)
- **SCNet:** 2,208 images (**22.5%**)
- **Semi-UIR:** 1,888 images (**19.3%**)

Under OOF evaluation, raw images attained 0.5808 mAP@0.5:0.95 (0.6873 per-image mAP), outperforming uniform SCNet (0.5785 full, 0.6859 per-image) and uniform Semi-UIR (0.5729 full, 0.6817 per-image). An OOF oracle reached 0.7164 per-image mAP (+0.0291 over raw).

C. Detection Results on RUOD Test Set

Table III reports the benchmark across all policies on the 4,200 RUOD test images using the unified detector.

1) Uniform Enhancement Harms Detection: Uniformly applying SCNet degrades mAP@0.5:0.95 by -0.0026 (-0.0031 per-image, CI $[-0.0049; -0.0013]$), while Semi-UIR lowers accuracy by -0.0071 (-0.0056 per-image, CI $[-0.0077; -0.0034]$). Both confidence intervals lie entirely below zero ($p_{\text{boot}} = 1.000$). Random selection similarly reduces accuracy (-0.0038).

2) Selector Safety and Baseline Parity: Uncalibrated selectors (`ce` and `reg`) also exhibit minor net degradation due to false-positive enhancement calls. In contrast, thresholded **Selector `ce`+τ** attains 0.6159 mAP@0.5:0.95 (+3.2% dataset gap closed), matching the raw baseline (0.6155) with 95% CI $[-0.0010; +0.0003]$ spanning zero ($p_{\text{boot}} = 0.851$).

3) Behavioral Diagnosis via Confusion Matrix: Table IV details the selector's validation performance and test confusion matrix. On test data, overall classification accuracy is 56.9% (below naive always-raw accuracy of 62.4%). Among 846 test images where SCNet truly improved mAP, the selector recalled only 83 (9.8%). For Semi-UIR, it recalled 48 of 734 images (6.5%).

The selector achieves baseline parity because the calibrated threshold ($\tau = 0.625$) enforces retention of the raw image for **85.5% of queries** (3,593 images). This defensive behavior protects the detector from degradation while executing in 7.7 ms.

4) Per-Class Breakdown: Table V presents per-class AP@0.5:0.95 across all 10 categories. The selector tracks the raw baseline on benthic classes (*holothurian, echinus, scallop*) while achieving its highest gain on *jellyfish* (0.6089 vs. 0.6026), where semi-transparent boundaries benefit from selective contrast recovery.

D. Dissecting the Oracle Upper Bound (The Noise Floor)

To investigate why learned selectors fail to capture oracle headroom, Table VI details the noise floor experiments on the 4,200 test images.

Key scientific findings:

1) **Spatial Reflection Exceeds UIE:** An oracle over raw and horizontally flipped raw images ($\{\text{orig}, \text{flip}\}$) attains **0.6300** dataset mAP—matching the 3-candidate UIE oracle. At the per-image level, the flip oracle gains +0.0207, representing **144%** of SCNet's gain (+0.0144).
2) **Mathematical Property of Extrema:** Because the flipped image underwent zero optical restoration, its gain stems entirely from detector coordinate perturbation and NMS sensitivity. The expected maximum of random variables $\mathbb{E}[\max(X_1, \ldots, X_K)]$ increases monotonically with K. Adding candidates (from 2 to 3 to 4) inevitably expands the oracle bound regardless of optical validity.
3) **Why Selectors Cannot Learn:** Because OOF training targets are largely governed by stochastic detector sensitivity rather than deterministic visual degradation patterns (correlation between flip benefit and SCNet benefit is 0.306), the target labels behave essentially as measurement noise. A convolutional selector inspecting pixels cannot learn to predict detector perturbation variance, demonstrating that mixed-set gains reported in prior literature [5] largely represent measurement noise.

E. Ensemble Fusion and Heuristic Baselines

Table VII evaluates practical alternatives that do not require ground-truth labels.

1) Failure of Confidence Heuristics: Selecting candidates by detector confidence (`conf_top5` or `conf_sum`) significantly lowers accuracy (-0.0025 and -0.0018 mAP). Enhancers accentuate high-frequency textures and edge gradients, inflating confidence on background false positives. High detector confidence does not correlate with true-positive precision.

2) Success of 4-View Geometric-Photometric WBF: Dual-view and three-view WBF configurations fail to outperform raw inputs. However, **4-view WBF combining geometric reflection and multi-enhancer views** reaches 0.6177 mAP@0.5:0.95 (+0.0023 dataset gain, +0.0032 per-image gain). Its 95% CI ($[+0.0014; +0.0051]$) is strictly positive, marking it as the sole inference technique achieving statistically verified improvements over raw images.

TABLE III
OBJECT DETECTION PERFORMANCE ON THE RUOD TEST SET (4,200 IMAGES) WITH YOLO11N.

Policy	mAP@0.5	mAP@0.5:0.95	Per-image mAP	Gap Closed (All / Per)	Δ vs. Raw [95% CI]	$p_{\text{boot}}(\Delta \leq 0)$	Selection (Orig / SC / Semi)
Always Raw (`orig`)	0.8468	0.6155	0.7186	0.0% / 0.0%	– (Baseline)	1.000	4200 / 0 / 0
Always SCNet	0.8449	0.6129	0.7154	-18.1% / -13.6%	-0.0031 [-0.0049; -0.0013]	1.000	0 / 4200 / 0
Always Semi-UIR	0.8422	0.6084	0.7130	-50.0% / -24.2%	-0.0056 [-0.0077; -0.0034]	1.000	0 / 0 / 4200
Random Selection	0.8444	0.6120	0.7148	-24.7% / -16.3%	-0.0038 [-0.0054; -0.0021]	1.000	1387 / 1401 / 1412
Selector `ce`	0.8472	0.6146	0.7173	-6.4% / -5.4%	-0.0012 [-0.0025; 0.0000]	0.970	2210 / 1101 / 889
Selector `ce`+τ	**0.8474**	**0.6159**	0.7182	**+3.2%** / -1.5%	**-0.0004 [-0.0010; +0.0003]**	0.851	**3593 / 319 / 288**
Selector `reg`	0.8457	0.6142	0.7182	-8.8% / -1.7%	-0.0004 [-0.0018; +0.0009]	0.734	2133 / 1373 / 694
Selector `reg`+τ	0.8462	0.6146	0.7183	-5.9% / -1.4%	-0.0003 [-0.0014; +0.0008]	0.745	3019 / 828 / 353
Oracle (Upper Bound)	**0.8524**	**0.6296**	**0.7417**	100.0% / 100.0%	+0.0231 [+0.0216; +0.0247]	0.000	2620 / 846 / 734

TABLE IV
VALIDATION PERFORMANCE ON 1,960 HELD-OUT IMAGES AND TEST CONFUSION MATRIX OF SELECTOR `CE`+τ ON 4,200 TEST IMAGES.

A. Validation Split Performance (1,960 images)

Variant	τ	Chosen-mAP	Gap Closed
Always Raw	–	0.6924	0.0%
`ce`	0	0.6923	-0.4%
`ce`+τ	**0.625**	**0.6931**	**+2.5%**
`reg`	0	0.6918	-2.4%
`reg`+τ	0.040	0.6931	+2.2%
Oracle	–	0.7202	100.0%

B. Test Confusion Matrix (Rows: True Oracle, Columns: Pred.)

Oracle Label	Pred. Raw	Pred. SCNet	Pred. Semi-UIR	Total
Raw (`orig`)	**2,259** (86.2%)	176 (6.7%)	185 (7.1%)	2,620
SCNet	708 (83.7%)	**83** (9.8%)	55 (6.5%)	846
Semi-UIR	626 (85.3%)	60 (8.2%)	**48** (6.5%)	734
Total Predicted	3,593 (85.5%)	319 (7.6%)	288 (6.9%)	4,200

TABLE V
PER-CLASS AP@0.5:0.95 ACROSS INPUT POLICIES ON RUOD TEST SET.

Class	Raw	SCNet	Semi-UIR	Selector `ce`+τ	Oracle
holothurian	0.4966	0.4910	0.4877	0.4966	0.5102
echinus	0.5305	0.5250	0.5187	0.5298	0.5446
scallop	0.4957	0.4941	0.4791	0.4958	0.5058
starfish	0.5508	0.5409	0.5431	0.5506	0.5652
fish	0.5198	0.5196	0.5181	0.5191	0.5337
corals	0.5386	0.5367	0.5338	0.5377	0.5519
diver	0.7456	0.7467	0.7397	0.7462	0.7638
cuttlefish	0.8360	0.8320	0.8315	0.8361	0.8487
turtle	0.8383	0.8390	0.8330	0.8384	0.8518
jellyfish	0.6026	0.6041	0.5991	**0.6089**	0.6208
Mean	**0.6155**	**0.6129**	**0.6084**	**0.6159**	**0.6296**

F. Multi-Version Detector Training as Data Augmentation

To evaluate whether UIE provides utility during detector training, Table VIII compares the single-domain baseline detector against the final multi-version detector evaluated on raw test images using the Ultralytics validator.

Training the detector across all three image versions (raw, SCNet, Semi-UIR) for 50 epochs does not harm raw image detection; on the contrary, it improves mAP@0.5:0.95 from 0.6037 to 0.6210 (+0.0173 absolute gain) and mAP@0.5 from 0.8446 to 0.8540 on raw test images. On the multi-version detector, per-class mAP@0.5:0.95 reaches 0.501 (*holothurian*), 0.532 (*echinus*), 0.501 (*scallop*), 0.555 (*starfish*), 0.527 (*fish*), 0.552 (*corals*), 0.751 (*diver*), 0.842 (*cuttlefish*), 0.843 (*turtle*), and 0.608 (*jellyfish*).

TABLE VI
ORACLE COMPARISON BETWEEN OPTICAL ENHANCERS AND HORIZONTAL REFLECTION NOISE ON RUOD TEST SET.

Oracle Candidate Set	Size	mAP@0.5:0.95	Δ Per-Image [95% CI]	Non-Raw %
{orig, orig rerun}	2	0.6159	+0.0005 [+0.0003; +0.0006]	2.3%
{orig, scnet}	2	0.6250	+0.0144 [+0.0133; +0.0154]	31.6%
{orig, semiuir}	2	0.6250	+0.0164 [+0.0150; +0.0179]	31.7%
{orig, flip} (Noise)	**2**	**0.6300**	**+0.0207** [+0.0193; +0.0221]	**38.5%**
Proposed UIE Oracle	3	0.6300	+0.0234 [+0.0219; +0.0250]	44.2%
{orig, flip, SC, Semi}	4	0.6386	+0.0353 [+0.0336; +0.0371]	58.6%

TABLE VII
MULTI-VIEW FUSION AND CONFIDENCE SELECTION ON RUOD TEST SET.

Method	mAP@0.5:0.95	Δ Full	Δ Per-Image [95% CI]
Raw Baseline	0.6155	–	–
WBF(orig, flip)	0.6145	-0.0009	-0.0002 [-0.0019; +0.0014]
WBF(orig, SCNet)	0.6085	-0.0070	-0.0050 [-0.0065; -0.0035]
WBF(orig, SCNet, Semi-UIR)	0.6111	-0.0044	-0.0025 [-0.0044; -0.0007]
WBF(orig, flip, SC, Semi)	**0.6177**	**+0.0023**	**+0.0032 [+0.0014; +0.0051]**
Confidence `conf_top5`	0.6129	-0.0025	-0.0021 [-0.0035; -0.0006]
Confidence `conf_sum`	0.6136	-0.0018	-0.0018 [-0.0034; -0.0003]

This confirms the central operational finding of our study: **UIE's primary utility for underwater object detection is not per-image selection at inference time, but offline multi-version data augmentation during training.** Incorporating synthetic water representations during training enriches feature representations and regularizes deep detectors against aquatic domain shifts.

VI. CONCLUSION

In this paper, we conducted a rigorous investigation into selective underwater image enhancement for object detection using YOLO11n on the RUOD benchmark. Our conclusions are:

1) **Uniform enhancement degrades detection:** Applying SCNet or Semi-UIR across the entire dataset lowers mAP@0.5:0.95 by -0.0026 and -0.0071 compared to raw inputs.
2) **Learned selectors offer computational safety:** A lightweight ResNet18 selector (7.7 ms) calibrated via validation thresholding ($\tau = 0.625$) matches raw baseline performance (0.6159 mAP@0.5:0.95, +3.2% gap closed) by defaulting to raw images in 85.5% of cases, preventing degradation.

TABLE VIII
COMPARISON OF YOLO11N BASELINE (TRAINED ON RAW ONLY) AND MULTI-VERSION DETECTOR EVALUATED ON RAW TEST IMAGES (ULTRALYTICS VALIDATOR).

Detector Model	Training Data	Epochs	mAP@0.5	mAP@0.5:0.95
YOLO11n Baseline (Raw)	9,800 raw images	50	0.8446	0.6037
YOLO11n Multi-Version	**29,400 images (3 versions)**	**50**	**0.8540**	**0.6210**
Difference (Δ)	+19,600 enhanced images	–	**+0.0094**	**+0.0173**

3) **The oracle upper bound is dominated by detector noise:** An unenhanced horizontal flip oracle accounts for 144% of SCNet's oracle gain (+0.0207 per-image mAP), proving that theoretical headroom is primarily detector sensitivity rather than learnable optical features.
4) **Confidence heuristics fail:** Selecting candidates by detector confidence degrades accuracy by up to -0.0025 due to artifact-induced false positives.
5) **Hybrid 4-view WBF reliably improves accuracy:** Combining spatial reflection and photometric UIE views via Weighted Boxes Fusion achieves statistically validated improvements (+0.0023 mAP@0.5:0.95, 95% CI $[+0.0014, +0.0051]$).
6) **Multi-version training functions as effective data augmentation:** Training the detector across all three image versions improves mAP@0.5:0.95 from 0.6037 to 0.6210 (+0.0173) over raw-only training, confirming that enhanced images provide valuable training-time data augmentation.

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